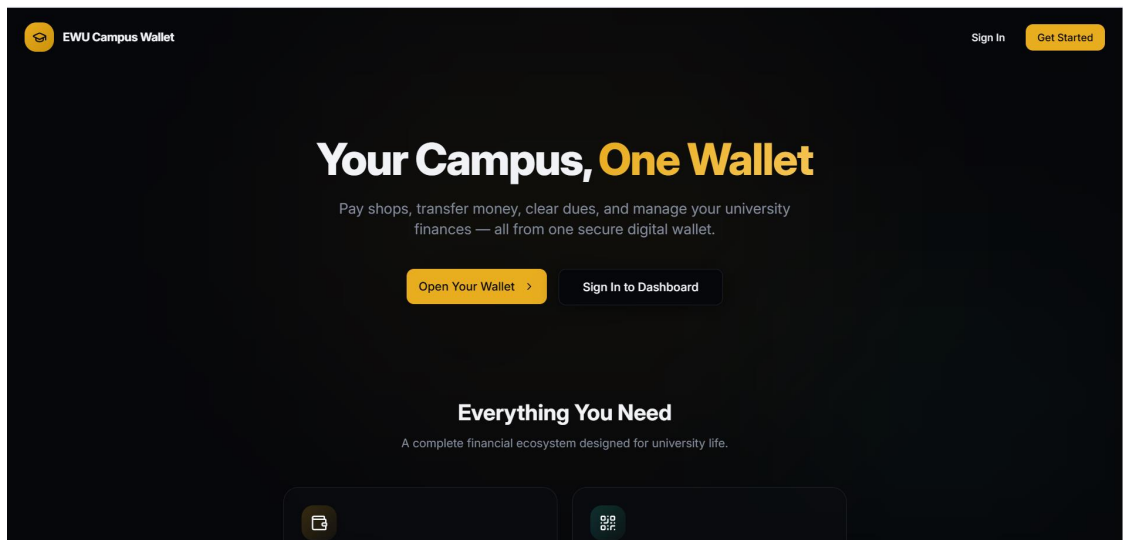


EWU Campus Wallet — Smart Campus

A multi-role digital super-app for East West University, unifying campus payments, dues & fines, and shop transactions into one wallet.

 **Live:** cse-mocha.vercel.app



Problem Area

University campuses typically run on fragmented, semi-manual systems for student finances: semester dues tracked on paper or spreadsheets, library and disciplinary fines settled in person, and canteen/shop purchases handled in cash only. This causes long queues, inconsistent record-keeping, no single view of what a student owes across departments, and heavy manual overhead for Admin, Accounts, and Library staff. **EWU Campus Wallet** solves this by giving every stakeholder — students, Admin Office, Library, Accounts Office, and campus shops — one digital platform to issue, pay, and track obligations in real time.

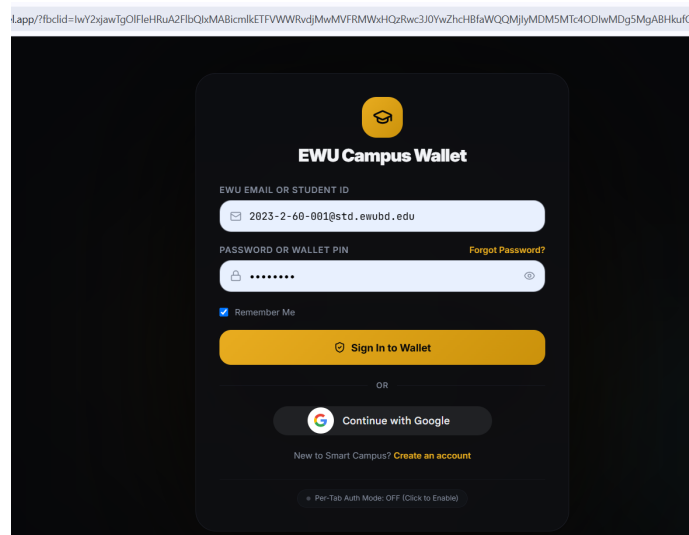
Project Summary

EWU Campus Wallet is a **multi-role university super-app** built around a central digital wallet. Students top up and spend from one wallet across campus — covering semester dues, library fines, disciplinary fines, and shop/canteen bills, with an option to pay later on shop purchases — while five separate role-based dashboards (Student, Admin Office, Library, Accounts Office, Shop) give staff the tools to issue, track, and resolve those obligations against the same wallet. The UI follows a premium, dark-themed fintech aesthetic, aiming to make a university utility feel as polished as a real banking app.

Features

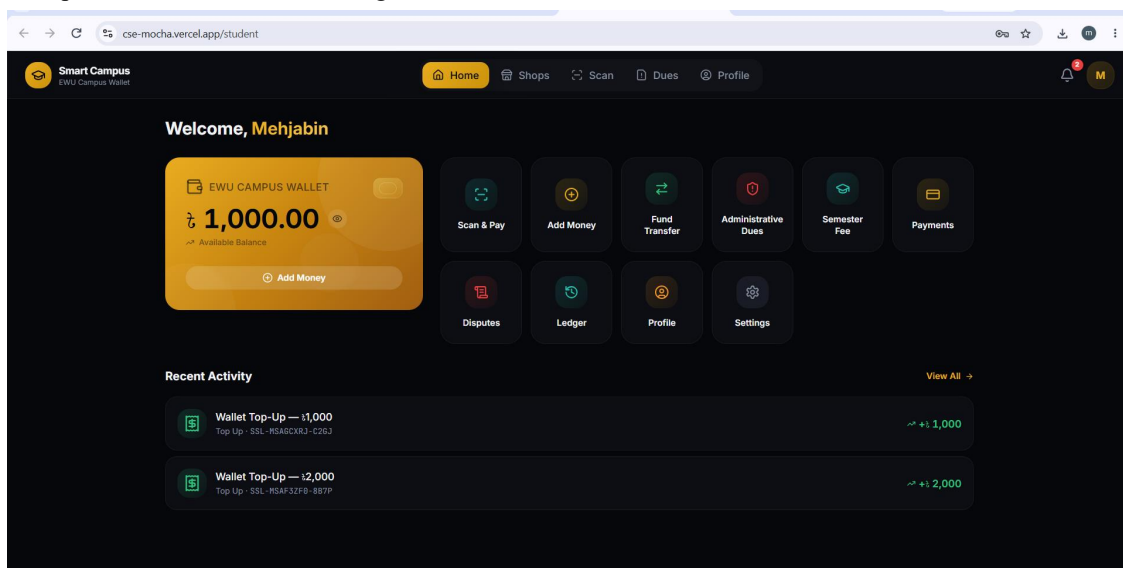
Authentication & Onboarding

- Sign in with an EWU email/Student ID and a password, or a wallet PIN
- One-tap **Google Sign-In**
- Optional **PIN-Tab quick-auth mode** for faster repeat logins
- Password recovery and new-account creation



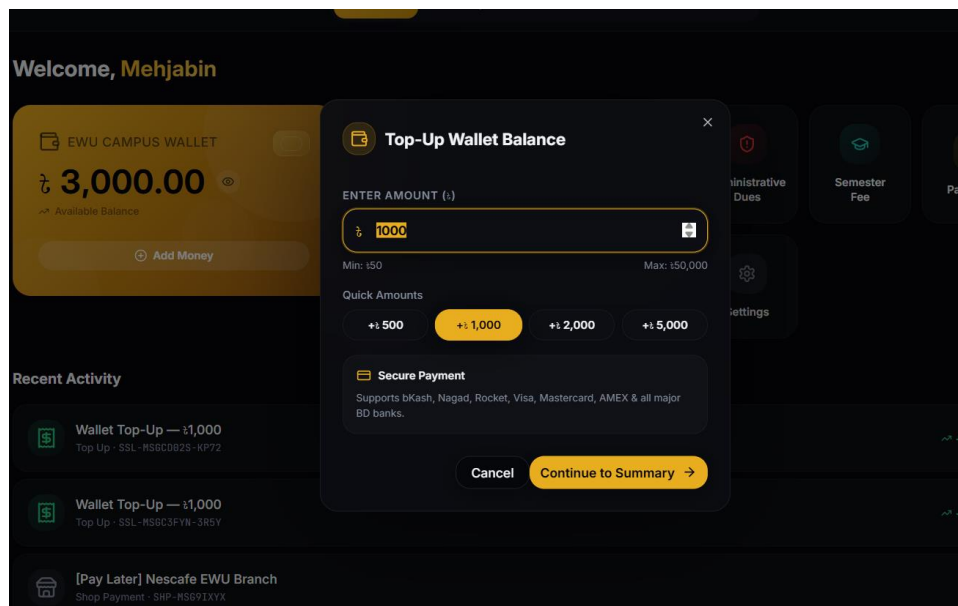
Digital Wallet & Home Dashboard

- One wallet per student with a live balance and a single **Add Money** action
- Quick-access grid to every module — Shop, Scan & Pay, Money Transfer, Fund, Semester Fee, Administrative Fee, Library Fine, Ledger, Profile, Settings
- A Recent Activity feed that unifies every transaction type — top-ups, fee payments, fines, and shop purchases — in one chronological list



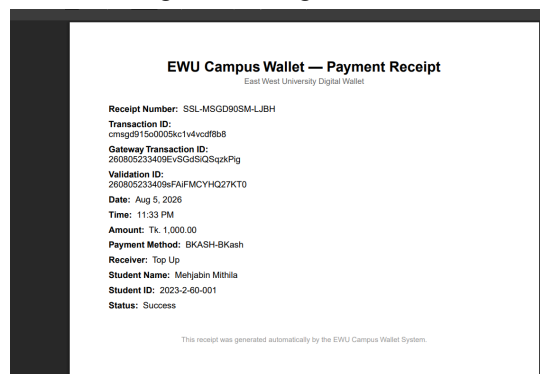
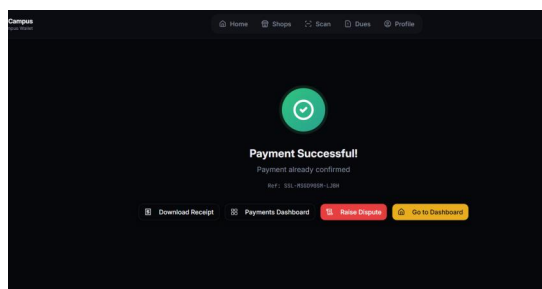
Wallet Top-Up

- Top up via **bKash, Nagad, Rocket, Visa, Mastercard, or AMEX**
- One-tap quick amounts (₹500 / ₹1,000 / ₹2,000 / ₹5,000) or a custom value
- A confirmation summary step before any money moves



Payments & Digital Receipts

- Real-time **Payment Successful** confirmation with a gateway reference number
- An auto-generated itemized receipt — transaction ID, gateway transaction ID, validation ID, date/time, amount, payment method, and student ID — downloadable any time
- **Raise Dispute** directly from a completed payment if something looks wrong



Pay Later

Students can **defer payment on a shop purchase** instead of paying at checkout. The purchase posts immediately to the wallet's Recent Activity, clearly tagged — e.g. **[Pay Later] Nescafe EWU Branch** — under its own Shop Payment reference, so a student always knows what's outstanding and to which shop. Unsettled Pay Later purchases accumulate into that shop's **Pending Settlement** balance, which the Admin Office later reconciles with the shop — giving students payment flexibility without slowing down the counter.

Dues & Fines Management

Three independent fine/fee sources, each with its own resolution path, unified under one **obligations** model so a student always sees a single picture of what they owe:

- **Admin Office (disciplinary) fines** — staff assign a fine with a reason, amount, and incident date; students can formally **appeal** it.
- **Library fines** — lost-book fees and counter payments, resolved directly with library staff, with a dedicated **Waivers** flow for cancelling or reducing a fine — no appeal step.
- **Accounts Office fees** — semester fees pushed in bulk to student accounts; adjustments are resolved directly — no appeal step.

←

Disciplinary Fines

+ Assign Fine Issued Fines Waivers

Search Student

Q 2023-2-60-001 Search

M Mehjabin Mithila
2023-2-60-001 - CSE - Batch 2023

Reason *

late fee

Amount (₹) * Incident Date

500 08/05/2026

Assign Fine

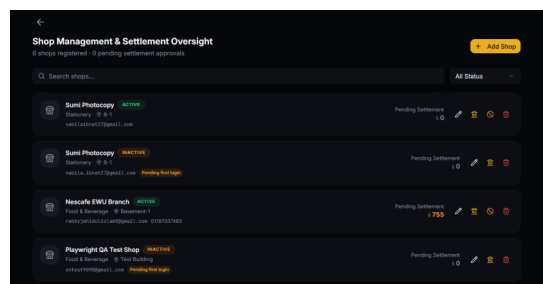
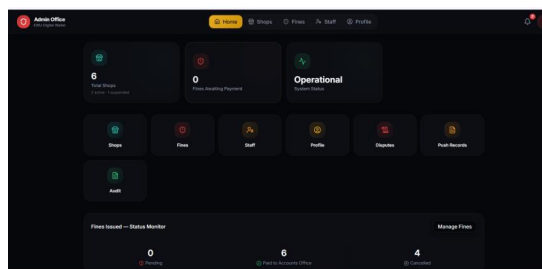
Campus Shop Payments

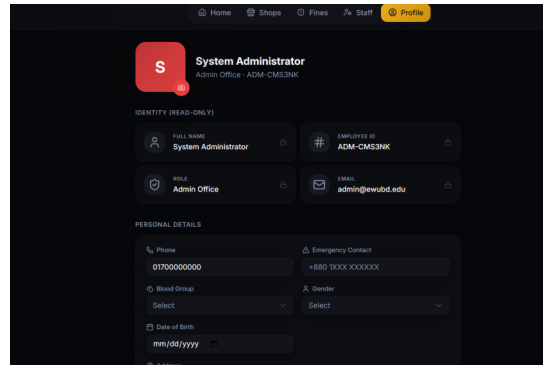
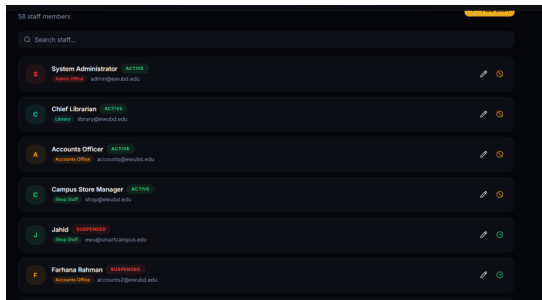
QR-code payments at campus shops and food counters, governed by a **prepaid-priority rule** — an active prepaid balance is spent down first, before the main wallet is touched — with the **Pay Later** option above for when a student wants to defer instead.

Admin Office Dashboard

The operational control center for the whole platform:

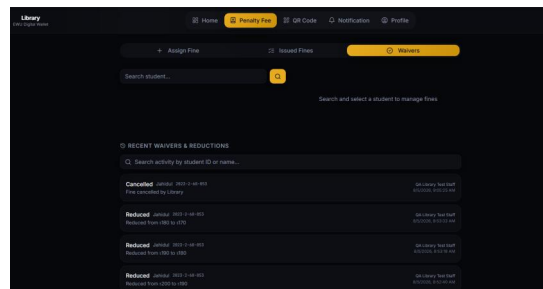
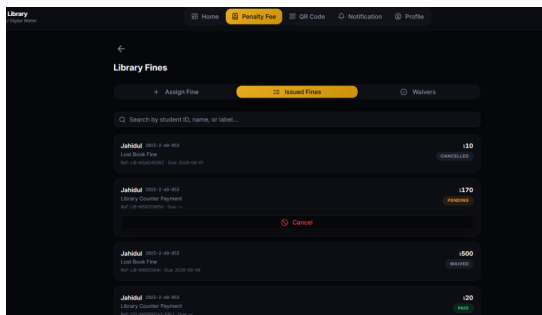
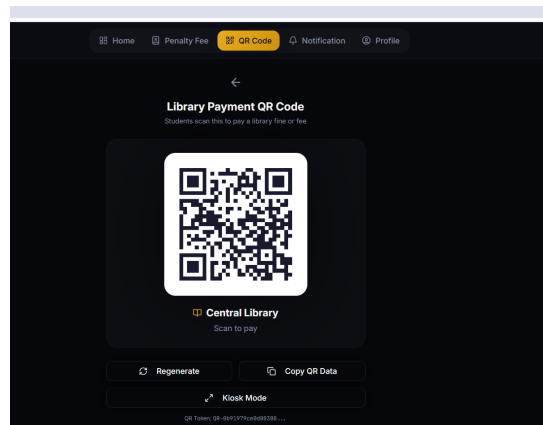
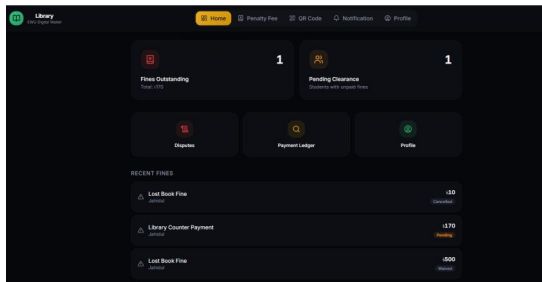
- At-a-glance stats — total shops, fines awaiting payment, system status — plus a Fines Issued status monitor (pending / paid to Accounts / cancelled)
- **Shop Management & Settlement Oversight** — activate or deactivate shops and clear each shop's pending settlement balance
- **Staff directory & provisioning** — every non-student account (Library, Accounts Office, Admin Office, Shop Staff) is created here and forced through a password reset on first login
- Staff profile pages — read-only core identity plus editable contact details
- Disputes and audit-trail access

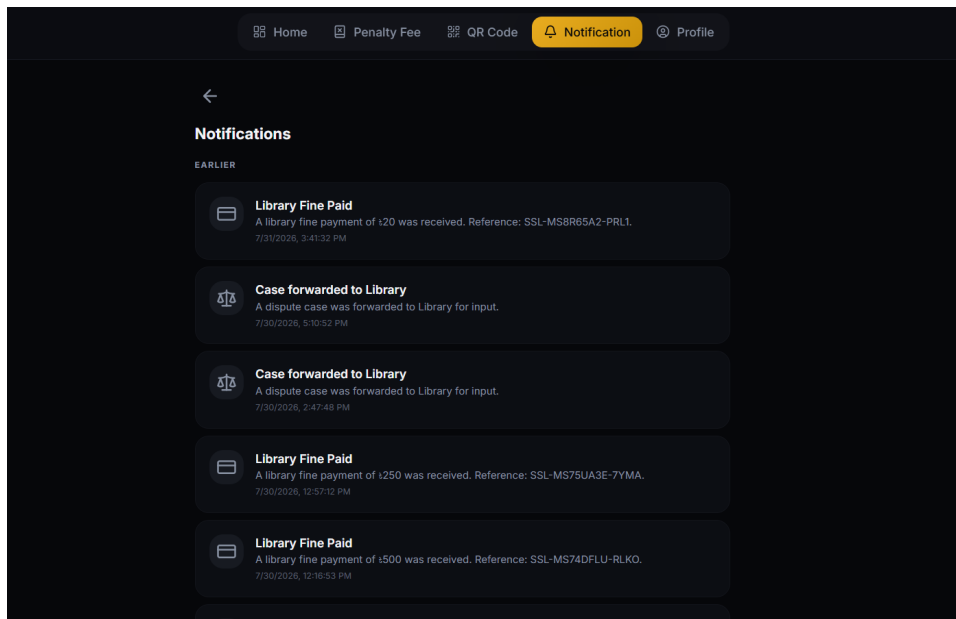




Library Dashboard

- Home overview of outstanding fines and pending clearances
- **Library Payment QR Code** — a regenerable QR code, plus a walk-up **Kiosk Mode**, so students can scan and pay a fine on the spot
- **Issued Fines** and **Waivers** workspaces for tracking and resolving or reducing a fine
- Live notifications when a fine is paid or a case is forwarded



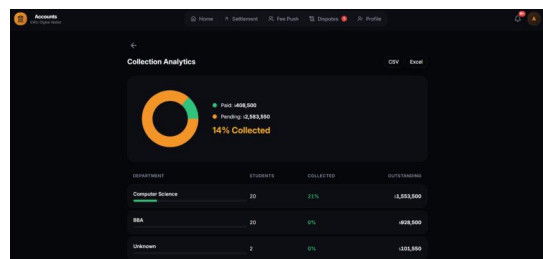


Accounts Office Dashboard

- **Fee Push Records** — bulk-push semester fees to the full student list, with per-student amount and status tracking
- **Collection Analytics** — a live paid-vs-pending breakdown by department, with a % collected rate
- **Financial Disputes** — review and resolve disputes students raised on a payment

The screenshot shows the 'Fee Push Records' dashboard with a table of fee push allocations. The table has columns: STUDENT ID, STUDENT NAME, DEPT / DEPARTMENT, LABEL / DESCRIPTION, AMOUNT, STATUS, and PAYER DATE.

STUDENT ID	STUDENT NAME	DEPT / DEPARTMENT	LABEL / DESCRIPTION	AMOUNT	STATUS	PAYER DATE
2023-2-08-024	Armed Mubarek	Computer Science	Summer 2026 Semester Fee	₦10,000	PENDING	Aug 1, 2026
2023-2-08-025	Armed Mubarek	Computer Science	Summer 2026 Semester Fee	₦10,000	PENDING	Aug 1, 2026
2023-2-08-026	Armed Mubarek	Computer Science	Summer 2026 Semester Fee	₦10,000	PENDING	Aug 1, 2026
2023-2-08-027	Armed Mubarek	Computer Science	Summer 2026 Semester Fee	₦10,000	PENDING	Aug 1, 2026
2023-2-08-028	Armed Mubarek	Computer Science	Summer 2026 Semester Fee	₦10,000	PENDING	Aug 1, 2026
2023-2-08-029	Armed Mubarek	Computer Science	Summer 2026 Semester Fee	₦10,000	PENDING	Aug 1, 2026
2023-2-08-030	Armed Mubarek	Computer Science	Summer 2026 Semester Fee	₦10,000	PENDING	Aug 1, 2026
2023-2-08-031	Armed Mubarek	Computer Science	Summer 2026 Semester Fee	₦10,000	PENDING	Aug 1, 2026
2023-2-08-032	Armed Mubarek	Computer Science	Summer 2026 Semester Fee	₦10,000	PENDING	Aug 1, 2026
2023-2-08-033	Armed Mubarek	Computer Science	Summer 2026 Semester Fee	₦10,000	PENDING	Aug 1, 2026
2023-2-08-034	Armed Mubarek	Computer Science	Summer 2026 Semester Fee	₦10,000	PENDING	Aug 1, 2026
2023-2-08-035	Armed Mubarek	Computer Science	Summer 2026 Semester Fee	₦10,000	PENDING	Aug 1, 2026
2023-2-08-036	Armed Mubarek	Computer Science	Summer 2026 Semester Fee	₦10,000	PENDING	Aug 1, 2026
2023-2-08-037	Armed Mubarek	Computer Science	Summer 2026 Semester Fee	₦10,000	PENDING	Aug 1, 2026
2023-2-08-038	Armed Mubarek	Computer Science	Summer 2026 Semester Fee	₦10,000	PENDING	Aug 1, 2026
2023-2-08-039	Armed Mubarek	Computer Science	Summer 2026 Semester Fee	₦10,000	PENDING	Aug 1, 2026
2023-2-08-040	Armed Mubarek	Computer Science	Summer 2026 Semester Fee	₦10,000	PENDING	Aug 1, 2026



The screenshot shows the 'Financial Disputes' dashboard. It includes a case management dashboard with various status counts and a list of disputes.

Open	Investigating	Waiting For Student	Waiting For Shop	Waiting For Library
0	0	1	0	0
Waiting For Admin	Resolved	Rejected	Refunded	
0	0	0	7	
0.5h Avg Resolution		1 SLA Overdue		₦1,040 Total Refunded

Search case #, category, student...

DSP-2026-000020	REFUNDED
Jahidul (2023-2-60-053)	
Accidental Payment - ₦1,500	
DSP-2026-000019	REFUNDED
Jahidul (2023-2-60-053)	
Waiting Resolution - 1.00	

Five Role-Based Dashboards

Role	Purpose
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Student	Wallet, top-up, dues/fines, Pay Later, appeals, shop payments
Admin Office	Issue/manage disciplinary fines, handle appeals, provision staff, oversee shops
Library	Fine imposition, QR/kiosk payment, waivers, notifications
Accounts Office	Fee push, collection analytics, dispute resolution
Shop	Per-shop QR payment acceptance & settlement

Tech Stack

- **Frontend:** React, TypeScript, Vite, Tailwind CSS
- **Backend:** Node.js, Express.js
- **Database:** PostgreSQL
- **Testing:** E2E test suite
- **Deployment:** Vercel

Demo Video

 <https://youtu.be/CLjAjvE5RWT?si=lQRDn7mRHJQyHFhF>